The Socratic Question Bank

HONR 46400 · Evidence-Driven Research

A reservoir of questions you can use word for word when you lead. Every question here is written to be said out loud by you, in class, exactly as it appears. Pull the ones your puzzle needs. The first half is organized by the SRL move you are making; the second half by the kind of research claim on the table.

A reminder on why these work. A good Socratic question hands the thinking back to the room. It cannot be answered with "yes" or "no," it does not hint at the answer you want, and it makes someone show their reasoning. When you feel the urge to explain, ask one of these instead.

Part A: Questions by move

Elicit prior beliefs

Get answers on the record before any evidence arrives.

1. "Before we look anything up, what is your gut answer, and how sure are you?"
2. "Hands up, who expects the effect to be positive? Negative? Zero? Keep your hand where it is, we are coming back to it."
3. "What would you have guessed a week ago, before this topic, and has that changed?"
4. "If you had to bet, which way would you bet, and what would make you switch sides?"

Force commitment

Lock in a written answer before the AI is opened.

5. "Write down your one-sentence answer now. We are not opening any tool until everyone has one."
6. "Commit to a number, even a rough one. A wrong number we can examine; a shrug we cannot."
7. "On the count of three, everyone say their answer at once so no one hides in the crowd."
8. "Before we ask the AI, what answer are you hoping it gives, and what answer would surprise you?"

Probe assumptions

Surface the condition a claim quietly depends on.

9. "What has to be true for that to hold?"
10. "You said that as if it were obvious. What are we taking for granted?"
11. "If I disagreed with you, which single assumption would I attack first?"
12. "Does that claim need the world to be a certain way? Describe that world."

Demand evidence

Ask what would actually support the claim.

13. "What evidence would we need to see to believe that?"
14. "Where would that data come from, and would it exist in the real world?"
15. "Is that a fact we can check, or an impression we are sharing?"
16. "If someone asked you to prove it, what is the first thing you would show them?"

Seek counterexamples

Test a claim by trying to break it.

17. "Give me a case where this would be completely false."
18. "Can you think of a situation the AI's answer did not consider?"
19. "If this rule is real, where does it stop working?"

20. “What is the strongest version of the opposite argument?”

Compare human versus AI

Put the room’s answer next to the machine’s.

21. “Here is what you committed to, and here is what the AI said. Where do they disagree, and who is right?”
22. “The AI agreed with us. Is that because we are right, or because it tends to agree?”
23. “If the AI and this room split, what evidence would settle it?”
24. “Read the AI’s answer and our answer side by side. Which one shows its reasoning, and which one just asserts?”

Expose AI overconfidence

Check whether the certainty is earned.

25. “The AI sounds sure. What is it actually basing that on?”
26. “It gave us a citation. Has anyone confirmed that source exists and says this?”
27. “Where in this answer does it tell us what it is uncertain about? If nowhere, is that honest?”
28. “If I asked the AI the opposite question, do you think it would sound just as confident?”

Connect to projects

Pull the skill into each person’s own research.

29. “Whose project has a claim shaped exactly like this one?”
30. “Take this move back to your own question. Where would it change what you can say?”
31. “If your project made this mistake, where would it show up in your poster?”
32. “Which of you will use this exact prompt on your own data this week?”

Close on a decision

End on a human commitment, defended.

33. “So what do we decide? Not what the AI decides, what do we?”
34. “Someone defend that decision against the hardest question in the room.”
35. “If you had to write this decision into your dossier tonight, what exactly would the sentence say?”
36. “What is the one thing we now believe that we did not believe 50 minutes ago, and how do we know it?”

Part B: Questions by research territory

For descriptive claims

A descriptive claim says what the world is or was, with no counterfactual.

37. “Is this describing what we observed, or claiming something beyond the data in front of us?”
38. “When we say ‘most,’ most of whom exactly? Who is inside that group and who is outside?”
39. “Would this same pattern show up if we measured it a different way?”
40. “Are we reporting a difference, or are we already hinting it was caused by something?”

For causal claims

A causal claim says an intervention would change an outcome.

41. “You used the word ‘effect.’ Is there an intervention here, or just two things moving together?”
42. “What else could explain this besides the cause you named?”
43. “If we could rewind and change only the one thing, would the outcome really move?”
44. “What is being compared to what? Where is the counterfactual, the version of the world that did not happen?”

For prediction claims

A prediction claim forecasts unseen cases against a baseline and a metric.

45. “Is this predicting new cases, or just describing the ones we already have?”
46. “What is the baseline we are beating? A prediction with no baseline is not a result yet.”
47. “Did any information from the future sneak into the training data? How would we catch that?”
48. “How would this do on cases it has never seen, and how do we know before we ship it?”

For measurement

Measurement is turning a concept into a number you can record.

49. “Are we measuring the thing we care about, or the thing that was easy to measure?”
50. “If two people measured this, would they get the same number? If not, why?”
51. “What does a value of zero actually mean here, and is that the real floor?”
52. “Whose definition of this concept are we using, and would someone reasonable define it differently?”

For sampling

Sampling is choosing which units you observe to say something about a larger group.

53. “Who is in the sample, and who could never have ended up in it?”
54. “If we claim this about the whole population, what buys us the right to reach past the people we actually observed?”
55. “Who is missing from this data, and would including them change the answer?”
56. “Is this sample big enough to notice the effect we are looking for, or could it miss it by chance?”

For experiments

An experiment assigns units to conditions to compare outcomes.

57. “How did units get into each group? Was it something that could bias the comparison?”
58. “What did the control group experience, and is it a fair comparison?”
59. “Could the people running or taking part in this have nudged the result without meaning to?”
60. “If we ran this exact experiment again tomorrow, how different could the result be?”

For robustness

Robustness is whether a finding survives reasonable changes to the analysis.

61. “If we dropped the most extreme data points, would the finding survive?”
62. “How many different ways did we analyze this before we got the answer we liked?”
63. “What is the most defensible alternative choice, and does the result hold under it?”
64. “Which single decision, if we changed it, would flip the conclusion?”

For communication

Communication is stating a finding with its uncertainty and limits.

65. “Say the finding with its uncertainty attached. What is the honest version?”
66. “What would a careful reader wrongly take away from this, and how do we stop them?”
67. “Which limitation, if we hid it, would most embarrass us later?”
68. “If the effect is real but small, does our headline still tell the truth?”

When you build your prep template, lift three or four of these and adapt them to your exact puzzle. The best SRLs do not read these verbatim forever. They learn the shape of a good question here, then write their own.